AKDENIZ UNIVERSITY · FACULTY OF ENGINEERING · DEPARTMENT OF COMPUTER ENGINEERING

CSE 491 — Project Proposal

Project title	Multi-Agent Medical AI
Group no.	3
Project supervisor	Hüseyin Gökhan Akçay
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Group members

#	Name and surname	Student no.	Role / responsibility in the project
1	Defne Demir	20230808010	Evaluation and Validation Engineer
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3	Hasan Saygılı	20220808041	Data Engineer

1. Summary

This project aims to develop an agentic vision-language system built around a backbone VLM, using renal nuclear imaging data from over 2,000 patients, including static and pinhole images. The system will investigate domain adaptation through fine-tuning, structured draft report generation, similar-case retrieval, and integration of these capabilities into an agentic workflow. Initial experiments will use the 7B model to compare fine-tuning configurations and select a suitable approach for the 32B model, subject to computational feasibility. If fine-tuning the larger model is impractical or performance is insufficient, the evaluated 7B model will serve as a fallback, and alternative medical vision-language models may be considered.

The final prototype will be assessed using predefined metrics for report generation, retrieval, and end-to-end system performance. Expected outputs include a dataset preparation pipeline, an integrated research prototype, and an evaluation report documenting results, limitations, and failure cases. The system is intended for research and preliminary clinical decision support, not autonomous diagnosis or replacement of medical professionals.

2. Problem Definition and Motivation

Vision-language models (VLMs) have the potential to support medical applications by combining image understanding with natural language processing. We believe they may be particularly useful in imaging-intensive medical fields, motivating our focus on nuclear medicine and renal nuclear imaging.

The project aims to develop an agentic vision-language system based on LingShu, a medical vision-language model, using a dataset containing more than 2000 patients with two types of renal nuclear imaging data, static and pinhole. The system will investigate three main capabilities: generating structured draft reports from imaging data, retrieving similar historical cases, and providing preliminary decision support. Rather than expecting a single model to perform every task, the proposed architecture will combine LingShu's image-text processing capabilities with separate modules for similar-case retrieval, output validation, and agentic workflow orchestration where appropriate.

If the prototype shows promising results, the architecture could be further developed and adapted to other imaging-intensive medical specialties. This would require additional domain-specific data, evaluation, and validation for each new application. The proposed system is intended as a research prototype for clinical decision support rather than a replacement for qualified medical professionals.

3. Objectives, Scope and Success Criteria

Objectives

The project aims to develop an agentic vision-language system for diagnostic assessment using LingShu. The main objectives are to:

- Assess and preprocess the available dataset. This dataset consists of around 2000 patients with pinhole and static imaging data.
- Conduct fine-tuning experiments and benchmark comparisons using the 7B model to identify the best-performing configuration.
- Apply the selected approach to the 32B model, subject to computational feasibility.
- Develop report-generation and similar-case retrieval modules.
- Integrate the modules into an agentic workflow and evaluate system performance.

Out of scope

The project will focus on a research prototype. The following are outside the current scope:

- Training a vision-language model from scratch.
- Replacing medical professionals or enabling autonomous clinical decision-making.
- Deploying the system for direct clinical use without expert validation.
- Guaranteeing successful fine-tuning of the 32B model regardless of computational limitations.

Success criteria

The project will be evaluated using established metrics and predefined test procedures:

- Report generation: ROUGE-L and BERTScore will be used to measure similarity between generated and reference reports. Where feasible, a clinically oriented metric such as RaTEScore will also be considered to assess medical content similarity. Clinical correctness and unsupported findings will be reviewed separately.
- Similar-case retrieval: Precision@5 and Mean Reciprocal Rank (MRR) will be used to evaluate the relevance and ranking of retrieved cases.
- Fine-tuning performance: The fine-tuned 7B model will be compared with its base version using the same validation data and evaluation metrics. Training time and peak GPU memory usage will also be recorded.
- System integration: Task completion rate and end-to-end latency will be measured using predefined test scenarios for the agentic workflow.
- Final evaluation: The selected model and integrated system will be evaluated on a held-out test set that is excluded from training and model selection. Results, limitations, and failure cases will be documented.

4. Similar Systems and Background

System / work	What it does	Limitation or gap your project addresses
LingShu [1]	A medical vision-language foundation model designed for multimodal medical understanding, reasoning, visual question answering, and medical report generation.	Our project will use LingShu as one component of a broader system, adapting it to renal nuclear imaging and integrating separate similar-case retrieval and agent-orchestration modules.
MediVLM [2]	A vision-language framework for generating radiology reports from medical images by combining visual feature extraction, image-text alignment, and report generation.	Its reported focus is general radiology report generation rather than our specific renal nuclear imaging. Our project will investigate adaptation to the available renal imaging data and integration with similar-case retrieval.
Observe, Align, and Enhance [3]	A retrieval-augmented radiology report-generation framework that retrieves similar images and associated reports, uses retrieved information to guide report generation, and refines the resulting report.	Our project will investigate a similar retrieval-and-reporting approach for renal nuclear imaging, integrated with LingShu and an agentic workflow. The suitability of this approach for our imaging types must be evaluated.

5. Preliminary Requirements

As the project progresses, new requirements may be identified. These requirements will be evaluated based on their relevance, feasibility, and alignment with the project objectives. Additional features may be incorporated when appropriate, provided that they fit within the available time and computational resources.

Functional requirements

ID	Requirement	Priority (must / should / could)
FR-1	The system shall use a fine-tuned LingShu model adapted to renal nuclear imaging data to identify and interpret relevant image features.	Must
FR-2	The system shall generate structured draft reports for supported imaging inputs using the selected LingShu model.	Must
FR-3	The system shall retrieve and display the top-k similar historical cases from the available dataset.	Must
FR-4	The system shall integrate report generation, similar-case retrieval, and output validation into an agentic workflow.	Must
FR-5	The agent shall select appropriate tools or modules based on the user's task.	Should
FR-6	The system could be extended to other medical imaging domains, provided that suitable datasets are available.	Could
FR-7	The system should provide a user interface to demonstrate report generation, similar-case retrieval, and the agentic workflow.	Should

Non-functional requirements

ID	Requirement	How it will be checked
NFR-1	The system shall protect patient data and restrict access to authorized users.	Review data-handling procedures, access restrictions, and anonymization practices.
NFR-2	The system shall process supported inputs within an acceptable response time.	Measure end-to-end inference latency on the available hardware.
NFR-3	The system shall use a modular architecture that allows individual components to be updated independently.	Verify that the model, retrieval, and agent modules can be tested and modified separately.
NFR-4	The system shall minimize hallucinations in generated reports.	Compare generated reports against available reference reports and identify unsupported findings.

6. Method and Technology Plan

The project will follow a two-stage model adaptation strategy. In the first stage, the 7B-parameter LingShu model will be used for baseline evaluation and preliminary fine-tuning experiments. This approach is intended to accelerate experimentation and enable efficient comparison of candidate training configurations, including suitable learning rates, training parameters, and adaptation techniques supported by the model.

The resulting configurations will be evaluated using predefined benchmark metrics on a validation set. Based on these results, the best-performing configuration will be selected for final implementation and fine-tuning on the 32B-parameter LingShu model.

In the second stage, the selected approach will be applied to the 32B model, and its performance will be evaluated on a held-out test set. If fine-tuning the 32B model proves infeasible because of hardware limitations, memory requirements, or project time constraints, the evaluated 7B model will be retained as a fallback.

The selected model will then be integrated with separate modules for structured report generation, similar-case retrieval, output validation, and agent orchestration, as appropriate. The final model choice and system capabilities will be documented together with the benchmark results and implementation limitations.

Layer / component	Technology or tool	Reason for the choice
Preliminary experimentation	LingShu 7B	Faster preliminary experimentation and benchmarking of candidate fine-tuning configurations.
Final model adaptation	LingShu 32B	Apply the selected fine-tuning approach to the primary model for final implementation.
Training framework	PyTorch and compatible LingShu tools	Support model adaptation and training experiments.
Evaluation	Task-specific benchmark metrics and validation scripts	Compare candidate configurations and evaluate the final model.
Similar-case retrieval	Embedding model and suitable retrieval index	Retrieve relevant historical cases independently of text generation.
Agent orchestration	Modular Python workflow or a suitable framework	Coordinate model inference, retrieval, and validation components.

7. Work Plan

WP	Work package and its output	Responsible member(s)	Weeks
WP1	Documentation for new features and project checkpoints, including bug tracking and fixes.	Defne, Zeynep, Hasan	1-12
WP2	Dataset inspection, cleaning, annotation analysis and preprocessing	Hasan	1-2
WP3	LingShu architecture review, 7B baseline evaluation and experimental setup	Defne	2-4
WP-4	Agentic system architecture design, module definition, tool selection, task routing, and workflow specification.	Zeynep	2-5
WP4	7B fine-tuning experiments, benchmark comparisons and selection of the best-performing configuration	Zeynep, Defne, Hasan	3–5
WP5	Fine-tuning the 32B model using the selected approach, subject to feasibility; otherwise finalize the fallback model.	Defne, Hasan	5–8

WP6	Similar-case retrieval, agent workflow integration and output validation	Zeynep	6–10
WP7	Integrated evaluation, final demonstration and project report	Defne, Zeynep, Hasan	11-12

8. Expected Outputs and Risks

The expected output is a prototype agentic system for renal nuclear imaging assessment, integrating LingShu with complementary software modules. The prototype is expected to include:

1. A documented and quality-assessed dataset preparation pipeline for the available renal nuclear imaging data.
2. A baseline evaluation and, if feasible, an adapted version of LingShu for selected image-understanding and report-generation tasks.
3. A module for generating structured draft reports from supported inputs.
4. A similar-case retrieval module that returns relevant historical cases from the available database.
5. An agentic workflow that integrates the implemented modules and presents outputs with available supporting case references.
6. An evaluation report documenting performance metrics, limitations, failure cases, and future improvement opportunities.
7. A source-code repository, setup instructions, and project documentation.

The prototype will be evaluated in a controlled research setting. It is not intended to replace qualified medical professionals or provide autonomous clinical diagnoses.

Risk	Likelihood	Impact	Mitigation plan
LingShu may not achieve satisfactory performance on renal nuclear imaging data.	Medium	High	If fine-tuning is infeasible or performance is unsatisfactory, investigate alternative medical vision-language backbones, such as HuatuoGPT-Vision[4], subject to hardware and compatibility constraints.
Fine-tuning the 32B model exceeds available computational resources or project time	High	High	Conduct preliminary experiments and benchmark comparisons on the 7B model, optimize the training configuration, and attempt 32B fine-tuning using the selected approach. If infeasible, retain the evaluated 7B model as a fallback.
Hallucinated or unsupported report findings	High	High	Compare outputs against reference reports, flag unsupported statements, use validation checks, and require expert review for clinical interpretation.
Poor performance on one or more image types	Medium	High	Evaluate each type separately, investigate failure patterns, and limit supported functionality to validated tasks.
Incorrect or irrelevant similar-case retrieval	Medium	High	Define relevance criteria, evaluate top-k results, and separate test cases from the retrieval index.
Inconsistent output formatting	Low	Low	Use a predefined output schema and validate generated reports before displaying them.

Selected evaluation metrics may not be suitable for the system.	Low	Medium	Review the applicability of the selected benchmarks to the available dataset and clinical tasks. If they prove unsuitable, identify and adopt alternative metrics that better reflect report quality, retrieval relevance, and system performance.

9. References

[1] LASA Team et al., "Lingshu: A generalist foundation model for unified multimodal medical understanding and reasoning," *arXiv preprint arXiv*:2506.07044, 2025. [Online]. Available: https://arxiv.org/abs/2506.07044

[2] D. Goswami, R. Subedi, and S. Chakraborty, "MediVLM: A vision language model for radiology report generation from medical images," in *Findings of the Association for Computational Linguistics: EMNLP 2025*, 2025, pp. 10287–10304. [Online]. Available: https://doi.org/10.18653/v1/2025.findings-emnlp.544

[3] K. Chen, X. Zhu, W. Zhang, S. Jiang, and Q. Chen, "Observe, align, and enhance: A hierarchical retrieval-augmented vision-language model for generating radiology reports" *Health Information Science and Systems, vol. 13, no. 1, art. no. 72, 2025.* [Online]. Available: https://doi.org/10.1007/s13755-025-00389-9

[4] J. Chen *et al.*, "Towards Injecting Medical Visual Knowledge into Multimodal LLMs at Scale," in *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP), 2024*, pp. 7346–7370. [Online]. Available: https://aclanthology.org/2024.emnlp-main.418/

Supervisor Approval

Supervisor name	Hüseyin Gökhan Akçay
Assessment	Approved X Approved with revisions ☐ Revise and resubmit ☐
Comments	Clear
Date / signature	